

Quiz (Carolina Reaper)

- Q1.** A cargo ship can transport 480 containers of goods. If each container holds 12 crates, how many crates can the ship transport in total, and what is the remainder when divided by 9?
(A) 5760 R 0
(B) 5760 R 3
(C) 5760 R 6
(D) 5760 R 9
(E) 5761 R 1
- Q2.** A truck driver needs to deliver 945 packages to a warehouse. If he delivers them in batches of 15, how many batches can he make, and what is the remainder?
(A) 63 R 0
(B) 63 R 2
(C) 63 R 5
(D) 63 R 10
(E) 62 R 15
- Q3.** A traffic light cycle lasts 120 seconds. How many times will it cycle in 18 minutes, using 2 as a divisor?
(A) 9 R 0
(B) 10 R 0
(C) 9 R 2
(D) 8 R 4
(E) 18 R 0
- Q4.** A logistics company has 270 parcels to ship. If they are shipped in boxes of 9, how many boxes can be filled, and what is the remainder?
(A) 29 R 3
(B) 30 R 0
(C) 30 R 3
(D) 29 R 6
(E) 30 R 6
- Q5.** A delivery service needs to divide 216 packages among 12 vans. What is the quotient and remainder when 216 is divided by 12?
(A) 18 R 0
(B) 17 R 6
(C) 18 R 4
(D) 18 R 6
(E) 19 R 6
- Q6.** A bus travels 630 miles in one day. If it travels the same route 5 times a week, how many miles does it travel per week using 5 as a divisor?
(A) 3150
(B) 3150 R 0
(C) 3151 R 1
(D) 3120 R 10
(E) 3152
- Q7.** A shipping company has 504 boxes of cargo. If they are packed in containers that hold 8 boxes each, how many containers can be filled, and what is the remainder?
(A) 63 R 0
(B) 63 R 1
(C) 63 R 4
(D) 62 R 8
(E) 63 R 8
- Q8.** A train travels 810 miles from one station to another. If it makes 5 stops along the way, and each stop takes the same amount of time, what is the remainder when 810 is divided by 5?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 5

Open Ended Questions (FRQ)

- Q9.** A delivery truck can carry 120 crates at a time. If 945 crates need to be delivered to a warehouse, how many trips must the truck make, and what is the remainder? Show your work and explain your reasoning.
- Q10.** A logistics company needs to pack 2400 boxes of goods into containers that hold 16 boxes each. What is the quotient and remainder when 2400 is divided by 16? Use long division to find the answer, and provide a brief explanation of the division algorithm used.

Chef's Tips

- Tip 1: Encourage students to use real-world examples to illustrate their reasoning.
- Tip 2: Remind students to show all work and explain their thought process for free-response questions.
- Tip 3: Emphasize the importance of accuracy and attention to detail in calculations.